

Two and a BillYan

Project Writeup for the CSCI1851 Final Project

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I. INTRODUCTION

Accurate skin lesion classification is a critical challenge where diagnostic precision directly influences patient outcomes. We propose an intermediate fusion architecture that integrates dermoscopic images with clinical metadata to classify skin lesions across seven categories in response to the Kaggle competition for CSCI1851. The framework applies transfer learning from a ConvNeXt-Base backbone for image feature extraction and a Multi-Layer Perceptron (MLP) to embed patient-specific variables into a latent space [1]. These feature vectors are processed via a joint-fusion head for final classification.

To optimize training we applied Linear Probing then Fine Tuning (LP-FT) with initial calibration of the tabular branch and classifier head with a frozen backbone, followed by end-to-end fine-tuning of the entire model [2]. We addressed class imbalance via inverse square-root frequency weighting and a robust image augmentation pipeline.

Our multimodal approach achieved a macro F1-score of 0.679 on the held out test set, significantly outperforming image-only (F1=0.65) and tabular-only (F1=0.17) baselines. Interpretability analysis revealed that biological sex is an important feature for Melanocytic nevus classification consistent with existing dermatological literature.

II. METHOD

We developed an intermediate fusion model that jointly processes dermoscopic images and clinical metadata to classify skin lesions into seven categories. The architecture consists of an image branch, a tabular branch, and a joint classifier head.

A. Image Branch

We use ConvNeXt-Base, a model pretrained on ImageNet as the image encoder [1], [3]. ConvNeXt-Base is a pure-convolutional architecture that modernizes the standard ResNet design by adopting improvements from vision transformers, such as depthwise convolutions and larger kernel sizes, to achieve state-of-the-art performance in computer vision tasks [4]. This module was selected over alternative CNN architectures as well as our own from-scratch CNN modules. We opted for a pre-trained CNN module we believed a model trained on the entirety of a large datasets such as ImageNet would have much more robust feature extraction capabilities compared to our own CNN modules that would be trained on 3,500 samples. Furthermore, the ConvNeXt-Base module

produced superior validation performance while maintaining reasonable computational cost/explainability relative to other image processing architectures such as vision transformers.

B. Tabular Branch

We use a simple multi-layer perceptron (MLP) to generate embeddings of the three given clinical features (age, sex, anatomical site) to pass into our joint classifier head.

C. Joint Classifier Head

The image and tabular embeddings are concatenated into a joint representation which is passed through another MLP for final classification. We chose intermediate fusion over early fusion, appending tabular and image features, which would eliminate opportunities to extract spatial features from images and late fusion, averaging independent predictions, which would learn independent patterns before final voting. By utilizing intermediate fusion, the model learns a joint latent space where tabular embeddings can contextualize high-level spatial representations in final class prediction.

D. Loss Function

To address the dataset's class imbalance, we use an inverse square root weighted cross-entropy loss which penalizes misclassifications of underrepresented lesion types more heavily. We also attempted to explore focal loss, however we found empirically worse results with this scheme – further hyperparameter tuning may be necessary [5]. Formally, the weight w_i for class i is normalized such that

$$w_i = \frac{C}{\sqrt{n_i} \cdot \sum_{j=1}^C \frac{1}{\sqrt{n_j}}}$$

where C is the total number of classes and n_i is the number of samples in each class. The weighted cross entropy loss per sample is thus defined as

$$L = - \sum_{c=1}^C w_c y_c \log(p_c)$$

III. EXPERIMENTAL SETUP

A. Dataset

We train and evaluate on the provided skin lesion dataset comprising 3,500 samples across seven diagnostic categories, paired with structured clinical variables. The dataset exhibits significant class imbalance as shown by figure 1.

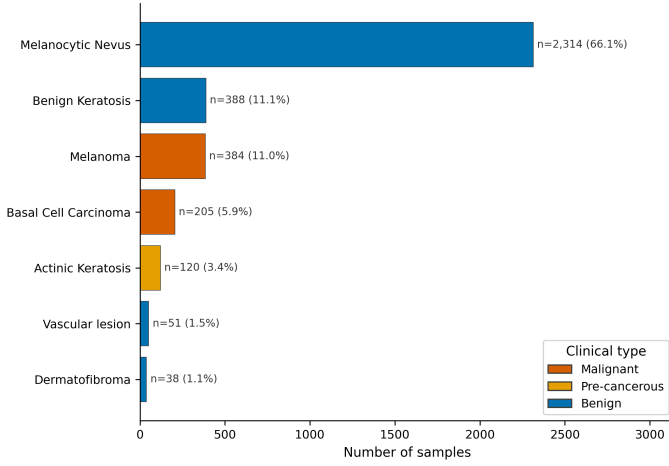


Fig. 1. Distribution of training dataset by class label

B. Pre-training and Augmentation

Images are resized to 224×224 and normalized using ImageNet mean and standard deviation. During training, we apply random horizontal and vertical flips, random rotation $\pm 15^\circ$, color jitter, and random erasing for regularization. Clinical features are also standardized prior to input.

C. Training Procedure

We adopt a two-phase staged training strategy inspired by the linear probe then fine-tuning scheme. First, the ConvNeXt-Base backbone is frozen and only the tabular MLP and fusion head are optimized allowing these components to be generally stabilized. Then, all parameters are jointly optimized using differential learning rates for the tabular branch and fusion head so that the pre-trained backbone can be fine tuned while the other parts of the model further trained. We use the AdamW optimizer with weight decay and apply early stopping with patience of 10 epochs. The dataset is an 80:20 stratified split into training and validation sets, with a batch size of 32.

D. Evaluation Metric

We report macro-F1 score as the primary metric which equally weights performance across all seven classes regardless of frequency. First $F1_i$ for each class i is calculated

$$F1_i = 2 \cdot \frac{\text{precision}_i \cdot \text{recall}_i}{\text{precision}_i + \text{recall}_i}$$

and then averaged across C classes as follows

$$F1_{\text{macro}} = \frac{1}{C} \sum_{i=1}^C F1_i$$

IV. RESULTS

Figure 2 illustrates the confusion matrix of our final model on our training data demonstrating that our design choices in the training are generally successful in limiting class imbalance issues in the distribution of predictions.

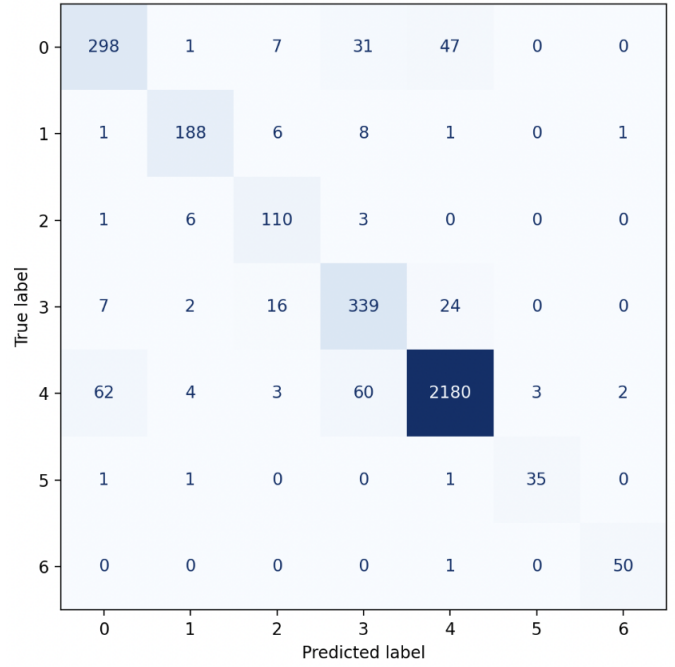


Fig. 2. Confusion matrix on training dataset

We further performed an ablation study where representation vectors of the respective image and tabular branches were zeroed out before being concatenated and passed to the joint classifier head to understand the contribution of the respective modalities.

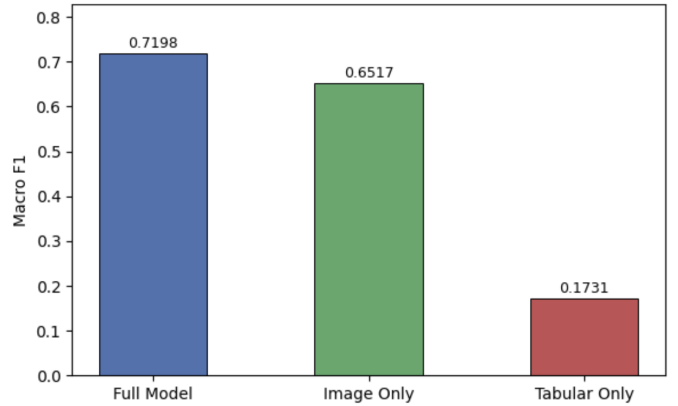


Fig. 3. Ablation study for modality specific predictions

The full multimodal model achieves a macro-F1 of 0.72 on the validation set, outperforming both the image only model (0.65) and the tabular baseline (0.17). The large gap between these two confirms that the dermoscopic images carry the majority of relevant signal, while improvement from image-only to the multimodal model shows that clinical metadata does provide meaningful complementary information that the model is sensitive to.

To understand input contributions to model predictions we apply a gradient attribution method. Formally, we develop a

contribution map by taking the magnitude of the gradients of the outputs y with respect to the inputs x [6].

$$M = \left| \frac{\partial y}{\partial x} \right|$$

As this is a gradient based approach, our analysis nicely decomposes into the respective model branches where a saliency map can be visualized below for the image modality and feature contributions can be observed for the tabular data.

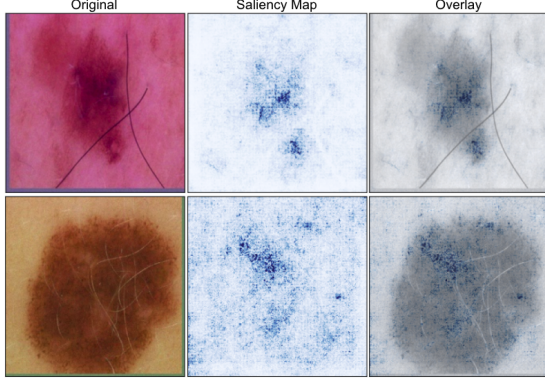


Fig. 4. Saliency maps calculated for from Melanocytic Nevus class

Figure 4 shows a raw input image, the calculated saliency map where darker blue pixels have larger gradient magnitudes, and the overlay shows the original picture and the overlay superimposed. Our visualizations reveal that the pixels from within the lesion contribute the most to model predictions, particularly pixels representing texture and boundary while assigning near-zero attribution to hair follicles and strands present. Hair is a common confounding artifact in dermatology images, and the model’s ability to disregard it suggests clinically relevant visual features have been learned.

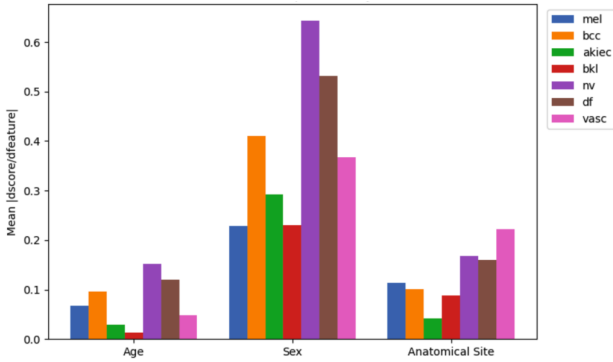


Fig. 5. Average gradient magnitude per clinical variable stratified by class

Visualizing the mean gradient attribution for each clinical variable stratified by target class below, we see that among the three variables, sex seems to be consistently the most influential. Furthermore, among the different target classes, we notice that sex contributes most to model predictions when the model is classifying samples from the nv (Melanocytic Nevus) class. This pattern for Melanocytic Nevus appears to be

biologically relevant where existing literature has documented that tumors have higher incidence on the head, neck and torso in males and on the legs in females [7], [8].

To further tease out the relationship between sex and anatomical location in our model, we utilize Integrated Gradients (IG), a path-based attribution method. IG attributes the prediction of a deep network to its input features by integrating the gradients of the output $F(x)$ along a linear path from a baseline x' to the input x [9]. Formally, the attribution for the i^{th} feature is defined as

$$IG_i(x) = (x_i - x'_i) \cdot \int_{\alpha=0}^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha$$

We designed a counterfactual experiment to determine if the model’s reliance on anatomical site is conditionally dependent on the patient’s sex. For a cohort of $N = 100$ samples correctly classified as Melanocytic Nevus (nv), we generated two distinct attribution maps by holding the image pixels and all non-sex tabular features constant while toggling the sex encoding in the baseline x' between 0 (Female) and 1 (Male). By defining the female encoding as the reference state, we isolate the interaction effect as the Δ Attribution

$$\Delta \text{Attribution} = \text{Attribution}_{\text{Male}} - \text{Attribution}_{\text{Female}}$$

This allows us to observe how the model’s internal anatomical site expectations of lesion distribution change when the demographic context is perturbed.

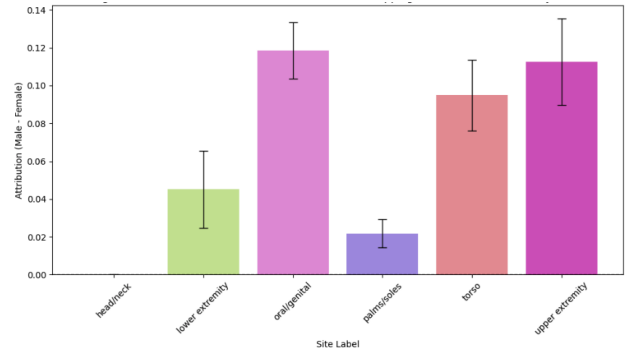


Fig. 6. Change in Feature Attribution of Anatomical Site When Swapping Sex Variable in Melanocytic Nevus Predictions

The counterfactual experiment reveals that the model’s spatial reasoning is not uniformly sensitive to patient sex. As illustrated in Figure 6, we observe significant positive Δ attribution for the torso, upper extremity, and oral/genital regions. This indicates that the model assigns higher diagnostic weight to these anatomical sites when a male context is provided compared to a female context. However, this is generally true across all anatomical sites other than the head/neck indicating that perhaps given that sex is male, there are more non-linear associations requiring the anatomical site as context for classification compared to classification when sex is female. The 0 difference in attribution for head/neck between males and females might indicate that the feature

does not contribute to predictions at all but further analysis is needed.

V. DISCUSSION

In this work we present a multimodal intermediate fusion model for skin lesion classification leveraging both dermoscopic images and clinical variables. By integrating a ConvNeXt-base image encoder with a tabular MLP feeding into a joint classifier head our model achieved a test macro-F1 of 0.662 on the held out Kaggle test set. Our findings demonstrate that while the image modality is the primary driver of diagnostic signal, the inclusion of patient-specific metadata via a LP-FT training strategy provides critical context that refines the model’s latent representations. Furthermore, our explainability analysis through saliency maps and integrated gradients confirmed that the architecture successfully learns clinically relevant features, such as lesion boundaries, while effectively disregarding artifacts like hair. Most notably, the counterfactual experiments highlight the model’s ability to capture complex interactions between demographic variables and anatomical sites, echoing established dermatological patterns. Ultimately, this research underscores the necessity of multimodal architectures and robust interpretability tools in developing reliable, biologically informed deep learning systems for computer-aided diagnosis in dermatology.

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VI. CONTRIBUTIONS

All team members contributed equally to project conception, presentation, and write up.

A. *Matias*

Matias developed initial late fusion models and spent time on feature engineering schemes that ended up not being needed based on our final submission.

B. *Stephen*

Stephen took lead on developing, implementing, and documenting the intermediate fusion model that ultimately became our final submission as well as explainability experiments.

C. *Bill*

Bill contributed extensively to model training efforts, hyperparameter tuning, and searching the modular pre-trained image backbones at our disposal for an optimal solution.